A Comparative Analysis of Cost Effectiveness of Cataract Surgery of Three Eye Care Providers in Kaduna State, North Central Nigeria.

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ABSTRACT

Cost-Effectiveness Analysis (CEA) estimates the costs and health gains of alternative interventions. Cataract surgery is the most common surgical procedure performed worldwide. This study aimed to determine and compare the cost effectiveness of cataract surgery by 3 different eye care providers in Kaduna State. This is a comparative study of the cost effectiveness of cataract surgery of 3 eye care providers in Kaduna State between 1st September 2021 to 17th November 2021. Adult patients aged 40 years and above without ocular co-morbidities were included. A semi-structured pre-tested interviewer-based questionnaire was administered pre-operatively and on post operative visits. Data obtained was entered and analysed using STATA version 15. A p value of <0.05 was considered statistically significant. There was a total of 165 patients with mean age of 64.54±9.91 and a male to female ratio of 1.03:1. The total direct and indirect cost of cataract surgery and from productivity loss was highest for National Eye Centre (NEC) at N263,855.27($637.33) and least for AMA foundation at N39,248.66 ($94.80). There was an improvement in the mean postoperative utility values compared to preoperative values. AMA foundation had the highest gain from 0.45 to 0.84 followed by NEC (0.47 to 0.85) and GSH (0.47 to 0.79). AMA foundation had the highest mean Quality Adjusted Life Years (QALYs) after cataract surgery. Cataract surgery was cost effective with AMA foundation, the most cost effective followed by NEC then GSH. The direct medical cost contributed to over half of the cost of cataract surgery compared to the direct non-medical cost.

Keywords: Cost effectiveness, Cataract surgery, Direct medical Cost, Indirect medical cost.

INTRODUCTION

Globally, at least 2.2 billion people have visual impairment, and of these, at least 1 billion people have visual impairment that could have been prevented or is yet to be addressed.[1] About 65.2 million people have visual impairment due to cataract. According to the Nigerian National Blindness and visual impairment survey, cataract is the commonest cause of severe visual impairment and blindness responsible for 45.3% and 43% respectively.[2] A fifth of Nigerian adults have cataract which increases with age, higher in females and those not literate.[3] Thus, cataract is a huge burden of visual impairment in the country. Treatment of cataract is by surgery and various surgical techniques are in use.[4] There is an on-going call for more productivity in eye health to improve the cataract surgical rates in populations especially in the developing countries where the surgical coverage has been reported to be low from several surveys despite the huge backlog of cataract blindness.[5] With a low cataract surgical coverage (people) of 4.0% and high couching coverage (people) of 18%, the main barrier to seeking cataract surgery was cost of the service in about 61% of persons with cataract.[6]

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Cost-effectiveness analysis can be undertaken in many ways, and there have been several attempts to develop methodological guidelines to make results more comparable. World Health Organization (WHO) has developed a standardized set of methods and tools that can be used to analyse the societal costs and impact on the health of the population of current and new interventions at the same time.[7]

In developing countries, 50% to 90% of blindness is due to cataract. Thus, the cost of the surgical procedure takes a main role as the number of surgeries increases. Cost-effectiveness here implies the economic analysis of relative costs and practical outcomes after cataract surgery. Cost effectiveness helps in interpreting the health care economic analysis in the specific field.[8] Cataract surgery is the most common procedure performed by the Ophthalmic surgeons and one of the most common surgical procedures worldwide. It is comparable in terms of cost-effectiveness to hip arthroplasty and is generally more cost-effective than either knee arthroplasty or defibrillation.[9] Studies have been done to show the outcome of cataract surgeries from Kaduna state but no known record on cost-effectiveness analysis.[10,11]

The results from this study can be used to compare the Cost-effectiveness of cataract surgery to the Cost-effectiveness of other disease interventions and also compare the Cost-effectiveness of cataract surgery in Kaduna State to other parts of the country. Thus, a health league table for policy makers and implementation of health interventions competing for limited resources can be generated.

MATERIALS AND METHODS

The study was carried out in 3 eye care facilities in Kaduna State namely National Eye Centre Kaduna (NEC), Aminu Musa Abdulsalam (AMA) Foundation (surgery carried out at Railway Hospital, Zaria) and Gambo Sawaba General Hospital (GSH) Zaria. It was a prospective comparative study of patients who had cataract surgery at 3 eye care facilities in Kaduna State, Nigeria between 1st September 2021 to 17th November 2021. Ethical approval for the study was obtained from the Research and Ethics committee of National Eye Centre Kaduna. Study permission was obtained from the Kaduna State Ministry of Health and AMA Foundation. The research was conducted in accordance with the tenets of the Helsinki declaration. Informed consent was obtained from all patients after proper counselling. Inclusion criteria include adult patients aged 40 years and above who were willing to undergo cataract surgery in at least one eye and agreed to come for follow up. Exclusions criteria include patients who were unwilling to participate in the study, those with poorly controlled diabetes mellitus, hypertension, and hyperlipidaemia and those with associated ocular co-morbidity e.g., trauma, corneal scar and glaucoma. The sample size was calculated from the formula for a cross-sectional study: $n = Z^2pq/d^2$, where n = sample size, Z = standard deviation (1.96), p = prevalence of the population with cataract blindness in those over 40 years (1.8%), q = 1-p and d = degree of precision (0.03). Consecutive adults who met the eligibility criteria were recruited for the study, until the required sample size of 165 was obtained. Sample size in each institution was based on the percentage of number of surgeries performed per year. So, the institution with the higher number of cataract surgeries had a higher ratio of the sample size. All patients had complete ocular examinations of both eyes. A semi-structured pre-tested interviewer-based questionnaire was administered along with the informed consent to the eligible patients both pre-operatively and on post-operative visits. The first section contained general patient demographic questions such as age, sex, marital status, level of education, number of accompanying persons, occupation and relationship along with number of pre- and post-operative hospital visits. A cost utilization survey section for the out-of-pocket cost expended while assessing cataract surgery. These costs included direct and indirect costs, direct costs being cost of cataract surgery, investigations, medications and any other additional charges related to hospital visits like transport fare, meals and accommodation while indirect costs were lost opportunities by the patient and their care givers. Data obtained was entered and analysed using STATA version 15. A p value of <0.05 was considered statistically significant. All study participants had their surgeries done by a consultant.

RESULTS

Majority of the patients were between the ages 61-80 years (61.80%) with a mean age of 64.54$\pm$9. 91years. National Eye center had a mean age of 66.61+8.16, Gambo Sawaba Hospital was 64.12+9.90 and AMA foundation was 60.33+13.90 which were statistically not significant. There was also no statistically significant difference in the gender distribution in the three groups as shown in Table 1 with the corresponding p values. More than half of the patients had Qur'anic education (58.20%) while only 12.10% and 10.30% had tertiary and primary education respectively. 81.20% of the patients were married and 18.80% were either divorced or widowed.

The mean direct medical cost **(Table 2)** per capita in National Eye Centre was N40,688 (USD98.28), N17,600 (USD42.51) in Gambo Sawaba General Hospital and zero cost in AMA foundation. The mean direct non-medical cost per capita was N13,418 (USD32.41) for National Eye Centre, N4,258 (USD10.28) for Gambo Sawaba General Hospital and around N766.66 (USD1.85) for patients of AMA foundation. The direct non-medical costs were for both the patient and accompanying person. The average travel distance was higher for National Eye Centre which was about 35km and lower for patients in Gambo Sawaba General Hospital which was about 10.5km. The cost of transportation accounted for about 58.51% of all the direct non-medical cost while accommodation accounted for just 14.58% because most accompanying persons stayed with their relatives in the hospital either on the wards or at the patient waiting area to save cost. Only patients from GSH were admitted after surgery, NEC and AMA foundation patients were operated as day cases. Analysis of indirect cost of patients and their accompanying persons **(Table 3)** for the three institutions showed a higher mean indirect cost (lost opportunity) for patients in National Eye Centre estimated to about N194,715.69(USD470.33) and N36,493.33(USD88.148) in AMA foundation while the accompanying person related cost was higher in Gambo Sawaba General Hospital.

Table 1: Socio demographic characteristics

Variables		NEC (n=51)	GSH (n=99)	AMA (n=15)	TOTAL (n=165)	P-value
Age	40-60	15(29.40%)	35(35.40%)	8(53.30%)	58(35.20%)	0.163
	61-80	33(64.70%)	63(63.60%)	6(40%)	102(61.80%)	
	>80	3(5.90%)	1(1%)	1(6.70%)	5(3%)	
Sex	Male	24(47.10%)	53(53.50%)	7(46.70%)	84(50.90%)	0.710
	Female	27(52.90%)	46(46.50%)	8(53.30%)	81(50.10%)	
Level of Education	Tertiary	11(21.60%)	9(9.10%)	0	20(12.10%)	0.167
	Secondary	7(13.70%)	9(9.10%)	1(6.70%)	17(10.30%)	
	Primary	8(15.70%)	13(10%)	3(20%)	24(14.50%)	
	Quranic	22(43.10%)	63(63.60%)	11(73.30%)	96(58.20%)	
	None	3(5.90%)	5(5.10%)	0	8(4.80%)	
Marital status	Married	40(78.40%)	82(82.80%)	12(80%)	134(81.20%)	
	Single	0	0	0	0	0.802
	Divorced /Separated/widowed	11(21.60%)	17(17.20%)	3(20%)	31(18.80%)	

The total direct and indirect cost gave the whole amount of funds spent in the hospital while accessing cataract surgery and from productivity loss. The highest amount was for National Eye Centre at N263,855.27($637.33) and least for AMA foundation at N39,248.66 ($94.80).

Table 2: Direct cost for cataract surgery

	NEC	GSH	AMA	Total	
Direct Medical Cost	Mean ± SD (Cost) n=51	Mean ± SD (Cost) n=99	Mean ± SD (Cost) n=15	Mean ± SD (Cost) n=165	ANOVA Statistics
Cataract Surgery cost	33000±0.00 (1683000)	16000±0.00 (1584000)	0	19800±9937.73 (3267000)	
Investigation cost	2480.39±1153.09 (126500)	500±0.00 (49500)	0	1066.67±1150.90 (76000)	F (183.719), P-value (0.001)
Medication cost	4021.57±1343.77 (205100)	600±0.00 (59400)	0	1603.03±1792.17 (264500)	F (391.57), P-value (0.001)
Additional procedures cost	1166.67±420.31 (59500)	500±0.00 (49500)	0	660.61±434.72 (109000)	F (203.209), P-value (0.001)
Total	**N2,074,100 ($5,009.9)**	**N1,742,400 ($4,208.69)**	**0**	**N3,816,500 ($9,218.59)**	
Direct Non-Medical Cost					
Transportation costs	7181.96±13017.72 (366280)	2809.09±2245.50 (278100)	635.33±430.36 (9530)	3963.09±7728.69 (653910)	F (7.462), P-value (0.001)
Feeding costs	3040.39±2019.04 (155060)	1449.69±1349.37 (143520)	131.33±191.98 (1970)	1821.52±1772.22 (300550)	F (27.997), P-value (0.001)
Accommodation costs	3196.08±980.19 (163000)	0	0	987.88±1577.23 (163000)	F (606.897), P-value (0.001)
Total	**684340.00**	**421620**	**11500**	**1117460.00**	
Grand Total Cost	**N2,758,440 ($6,662.89)**	**N2,164,020 ($5,227.10)**	**N11,500 ($27.77)**	**N4,933,960 ($11,917.77)**	

There was an improvement in the mean postoperative utility values **(Table 4)** as compared to the preoperative values with AMA foundation having the highest gain from 0.45 to 0.84 followed by NEC with an improvement from 0.47 to 0.85 and then GSH with improvement from 0.47 to 0.79. This shows the effectiveness was highest in AMA foundation then followed by NEC.

Gain in Quality Adjusted Life Years (QALYs) after cataract surgery was calculated for the three different institutions based on outcome **(Table 5**). AMA foundation had the highest mean QALYs.

DISCUSSION

This was a hospital based comparative study that determined the cost effectiveness of cataract surgery in 3 healthcare facilities in Kaduna State namely National Eye Centre Kaduna, Gambo Sawaba General Hospital Zaria and AMA foundation. In this study, there were slightly more males than females (M: F= 1.03:1) but not statistically significant. The results in this study are similar to a study conducted by Oladigbolu et.al in this environment and also another study in Kwara State by Adepoju et al where the M: F were 1:0.9 and 1:1.2 respectively indicating a bridging in the gap in the uptake of cataract surgery by females in this environment and dispels the fear of gender inequality to cataract surgery[10,14] and contrary to the what the Nigerian National blindness survey found.[13]

Table 3: Indirect cost due to cataract (Lost opportunity)

	NEC	GSH	AMA	TOTAL	
	Mean ± SD (Cost) n=51	Mean ± SD (Cost) n=99	Mean ± SD (Cost) n=15	Mean ± SD (Cost) n=165	ANOVA Statistics
Patient-related cost	194715.69±879750.13 (9930500)	24463.23±151171 (2421860)	36493.33±88845.61 (547400)	78180.36±506376.59 (12899760)	F (1.982), P-value (0.141)
Accompanying person-related cost	35747.06±131758.89 (1823100)	46502.65±254592.90 (4603762)	1988.67±2388.79 (29830)	39131.47±210210.07 (6456692)	F (0.299), P-value (0.742)
Total cost	**N11,753,600 ($28,390.33)**	**N7,025,622 ($16,970.10)**	**N577,230 ($1,394.27)**	**N19,356,452 ($46,754.72)**	

Table 4: Preop-Utility & Postop-Utility values

	NEC	GSH	AMA	TOTAL
	Mean ± SD n=51	Mean ± SD n=99	Mean ± SD n=15	Mean ± SD n=165
Pre-op Utility	0.47±0.11 (23.98)	0.47±0.10 (46.25)	0.45±0.13 (6.72)	0.47±0.10 (76.95)
Post-op-Utility	0.85±0.12 (43.51)	0.79±0.17 (78.68)	0.84±0.14 (12.66)	0.81±0.16 (134.85)

Table 5: QALYs gained based on probabilities of outcomes and cost

	NEC (n=51)	GSH (n=99)	AMA (n=15)
Probabilities of good outcome	0.86	0.71	0.80
Probabilities of moderate outcome	0.098	0.14	0.13
Probabilities of poor outcome	0.039	0.15	0.07
Cost for good outcome [mean]	**N1,789,419 (USD 4,322.26) [USD 84.75]**	**N1,466,533 (USD 3,542.35) [USD 35.78]**	**N1,659,280 (USD 4,007.92) [USD 276.19]**
Cost for moderate outcome [mean]	**N203,343 (USD 491.16) [USD 9.63]**	**N293,307 (USD 708.47) [USD 7.15]**	**N276,546 (USD 667.98) [USD 44.50]**
Cost for poor outcome [mean]	**N81,337 (USD 196.46) [USD 3.85]**	**N314,257 (USD 759.07) [USD 7.66]**	**N138,273 (USD 333.99) [USD 22.26]**
QALYs gained for good outcome	5.31	7.95	11.72
QALYs gained for moderate outcome	4.36	6.31	8.58
QALYs gained for poor outcome	2.61	4.17	6.76

Key: Good outcome =6/6-6/12
Moderate outcome =<6/12-6/60
Poor outcome =<6/60

Majority of the patients were within the age range of 61-80 years with a mean range of 64.54±9.91 years which is similar to the findings in Ibadan, Oyo State, Nigeria[15] with a mean of 65.8±8.46 years and 66.1 years at Onitsha by Nwosu et al.[16] This age distribution is in conformity with what is seen in cataract patients in Nigeria.[15] About 81.20% of the patients were married and only 18.80% where either divorced or widowed in this study as it is expected for this age range. It was found in this study that more than half (58.20%) of the patients did not have a formal education and 4.80% had no any form of education while 36.90% had some level of formal education which is similar to a study by Olawoye et.al.[15] The predominant occupation of the patients was found to be farming and trading which correlates to the level of education of the patients and was similar to what was seen in other studies in parts of Nigeria.[15]

The mean direct medical cost for cataract surgery was N40,688 (USD98.28) in National Eye Centre which was higher than the mean direct medical cost found in Gambo Sawaba General Hospital at N17,600 (USD42.51) and at no cost for AMA foundation. The direct non-medical cost was also found to be higher in National Eye Centre N13,418 (USD32.41) than in Gambo Sawaba General Hospital N4,258 (USD10.28) and AMA foundation N766.66 (USD1.85) because no accommodation was required in theses 2 centres. In general, the mean total direct medical cost was found to be 3 times higher than the direct non-medical cost in National Eye Centre and 4 times higher in Gambo Sawaba General Hospital. There was no direct medical cost in AMA foundation since patients had cataract surgery at no cost in this facility, however non direct medical cost was still incurred. The results in this study are contrary to a study carried out in Zamfara State where the direct non-medical cost doubled the total direct medical cost of cataract surgery.[17]

The cost effectiveness ratio per capita was N164,909.54($398.33) for National Eye Centre, N70,375($169.99) for Gambo Sawaba General Hospital and N23,830.39 ($57.56) for AMA foundation. This was calculated by dividing the total cost of cataract treatment per capita with the health outcomes which is the VA/Effectiveness. Cost effectiveness ratio can be thought of as the price at which an intervention produces health gains, that price must be compared with a benchmark to determine whether the intervention is cost effective. Cost effectiveness ratio exceeding the threshold are considered unfavourable. An intervention is considered very cost effective if its cost effectiveness ratio is less or equal to 1 GDP (gross domestic product)/per capita/ DALY, considered cost effective if its cost effectiveness ratio ranges from 1-3, and not cost effective if it is greater than three GDP/capita/DALY.[18]

GDP is the per capita income of Nigeria and measures the average income earned per person in a given area in a specified year. It is calculated by dividing the average total income of a nation by its total population. The GDP for Nigeria in 2021 was $2,360.00. The cost effectiveness ratio in this study was considerably lower than the GDP in all the three facilities and thus considered to be very cost effective. Lansingh et al conducted a cost effectiveness analysis of cataract surgery in nine countries around the world and found that in all the countries the cost effectiveness ratio was much lower than the GDP per capita thus making cataract surgery very cost-effective in all the locations.[9] Effectiveness derived from the visual outcome was highest for AMA foundation and with the least cost of N23,830.39 ($57.56), this made it the most cost-effective facility for cataract surgery. In the Zambian study the cost effectiveness ratio per QALY gained was $259 which was lower than the cost for National Eye Centre but higher than the cost effectiveness ratio of AMA foundation and Gambo General Sawaba Hospital.[19] A study by Aribaba in Lagos Nigeria[20] found that cataract surgery costs were surprisingly higher when compared to other developing African countries and was comparable to those of Busbee et al.[12]

CONCLUSION

Cataract surgery was found to be cost effective in this study. AMA foundation was the most cost effective followed by NEC then GSH. The direct medical cost contributed to a higher percentage of the economic burden of cataract surgery compared to the direct non-medical cost.

CONFLICTS OF INTEREST

None

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